



IT-SIMPLERA INSTITUTE

Network Administration Internship Program

WEEK 04 TASK REPORT

Enterprise Network Security and Access Control

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1. Introduction / Assignment Overview

This report explains the work done in Week 4 of the Network Administration Internship Program at IT-Simplera Institute. In Week 3, a working multi-department network was built. In Week 4, the goal was to make that same network safer by adding several security features that are used in real companies.

Access Control Lists (ACLs) were used to control which department is allowed to talk to which other department. SSH was set up so that only the IT team can safely connect to the router from a distance, instead of using the unsafe Telnet protocol. On top of that, several switch-level (Layer 2) security features were added: Port Security, DHCP Snooping, BPDU Guard, Root Guard, and PortFast. Together, these features protect the network from common mistakes and simple attacks, such as someone plugging in an unknown device or a rogue switch trying to take over the network.

Objectives

- Set up and test Standard, Extended, and Named Access Control Lists (ACLs) to control traffic between the HR, Finance, and IT departments.
- Set up SSH so the router can be managed safely from a distance, and restrict SSH access to the IT department only.
- Add Port Security so that only known computers (by MAC address) can use each switch port.
- Add DHCP Snooping so that a fake or rogue DHCP server on the network cannot hand out wrong IP addresses.
- Add PortFast and BPDU Guard on the ports connected to computers, so they connect quickly and are protected from accidental switching loops.
- Add Root Guard on the link between the two switches, so that no switch can wrongly become the root of the network, whether by accident or by an attack.
- Test every feature, and record the problems that came up during testing along with how each one was solved.

Tools Used

- GNS3 – used to build and run the virtual network (router and switches).
- PuTTY – used to open SSH sessions to the router for testing.

2. Network Design

2.1 Topology Overview

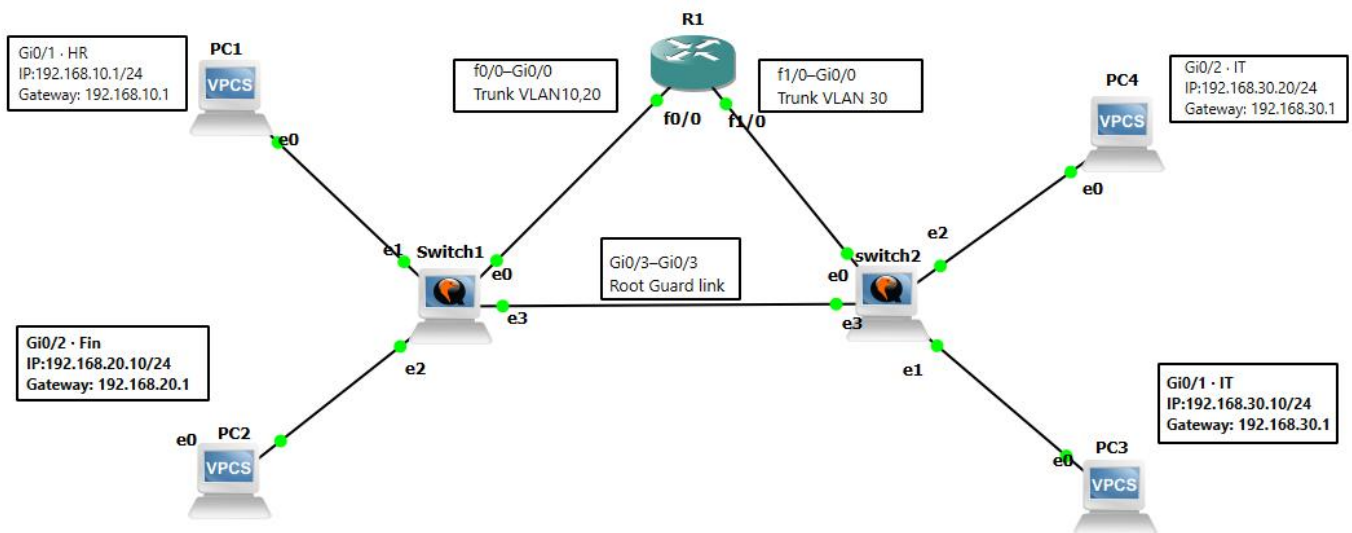
The Week 4 network is made up of one router (R1) and two switches (SW1 and SW2). Three separate departments were created using VLANs (Virtual LANs), which is a way of splitting one physical network into several logically separate networks:

- VLAN 10 – HR department, connected to switch SW1, used by PC1.
- VLAN 20 – Finance department, connected to switch SW1, used by PC2.
- VLAN 30 – IT department, connected to switch SW2, used by PC3 and PC4.

Router R1 is connected to both switches using trunk links. A trunk link is a single cable that can carry traffic for more than one VLAN at the same time, using small "sub-interfaces" on the router one sub-interface per VLAN. This lets one router serve three separate departments through just two physical cables.

A direct link was also added between SW1 and SW2, carrying only VLAN 1. This extra link does not carry any department traffic it exists only so that the Root Guard feature (explained later in this report) could be properly demonstrated, since Root Guard needs a direct switch-to-switch connection to be meaningful.

2.2 Topology Diagram



2.3 Device Roles Summary

Device	Role	VLAN(s)	Connected to
R1	Router inter-VLAN routing, SSH server, ACL enforcement point	10, 20, 30	SW1, SW2
SW1	Access switch	10 (HR), 20 (Finance)	R1, PC1, PC2, SW2
SW2	Access switch	30 (IT)	R1, PC3, PC4, SW1
PC1	HR host	10	SW1 – Gi0/1
PC2	Finance host	20	SW1 – Gi0/2
PC3	IT host	30	SW2 – Gi0/1
PC4	IT host	30	SW2 – Gi0/2

3. IP Addressing Table

Each department was given its own /24 subnet. The router's sub-interface address acts as the default gateway for that department.

Network	VLAN	Subnet	Gateway	Host IP addresses
HR-LAN	10	192.168.10.0/24	192.168.10.1	PC1: 192.168.10.10
Finance-LAN	20	192.168.20.0/24	192.168.20.1	PC2: 192.168.20.10
IT-LAN	30	192.168.30.0/24	192.168.30.1	PC3: 192.168.30.10, PC4: 192.168.30.20

Router Sub-interfaces

```
interface FastEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface FastEthernet0/0.20
```

```
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
interface FastEthernet1/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
```

```
R1#enable
R1#config te
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface fastethernet0/0
R1(config-if)#no shutdown
R1(config-if)#interface fastethernet1/0
R1(config-if)#no shutdown
R1(config-if)#
*Jul 24 12:55:13.615: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to u
*Jul 24 12:55:14.615: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1
0, changed state to up
R1(config-if)#exit
R1(config)#interface fastethernet0/0 .10
      ^
% Invalid input detected at '^' marker.

R1(config)#interface fastethernet0/0.10
R1(config-subif)#encapsulation dot1q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config-subif)#exit
R1(config)#interface fastethernet0/0.20
R1(config-subif)#encapsulation dot1q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
R1(config-subif)#exit
R1(config)#interface fastethernet1/0.30
R1(config-subif)#encapsulation dot1q 30
R1(config-subif)#ip address 192.168.30.1 255.255.255.0
R1(config-subif)#exit
R1(config)#end
```

4. Security Implementation

4.1 Standard ACL

What is a Standard ACL and why use it?

A Standard ACL is the simplest type of access list on a Cisco router. It can only check where traffic is coming FROM (the source IP address). It cannot check where the traffic is going or what kind of traffic it is (web, ping, email, and so on). Because it cannot be specific about the destination, best practice is to place a Standard ACL as close as possible to the destination network, so it does not accidentally block traffic meant for other places.

Purpose in this assignment: the HR department should not be able to reach the Finance department at all, to keep Finance's information private.

Configuration

```
access-list 10 deny 192.168.10.0 0.0.0.255
access-list 10 permit any
interface FastEthernet0/0.20
ip access-group 10 out
```

Verification

PC1 (HR) was tested against the Finance gateway and against the IT network to confirm that only Finance traffic is blocked.

Standard ACL configuration on R1

```

R1#enable
R1#config te
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 10 deny 192.168.10.0 0.0.0.255
R1(config)#access-list 10 permit any
R1(config)#interface fastethernet0/0.20
R1(config-subif)#ip access-group 10 out
R1(config-subif)#exit
R1(config)#end
R1#
*Jul 24 13:13:33.495: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#

```

PC1 ping to Finance (blocked) and to IT.

```

PC1> ping 192.168.20.10

*192.168.10.1 icmp_seq=1 ttl=255 time=46.454 ms (ICMP type:3, code:13, Communication
administratively prohibited)
*192.168.10.1 icmp_seq=2 ttl=255 time=35.880 ms (ICMP type:3, code:13, Communication
administratively prohibited)
*192.168.10.1 icmp_seq=3 ttl=255 time=8.089 ms (ICMP type:3, code:13, Communication
administratively prohibited)
*192.168.10.1 icmp_seq=4 ttl=255 time=5.163 ms (ICMP type:3, code:13, Communication
administratively prohibited)
*192.168.10.1 icmp_seq=5 ttl=255 time=8.645 ms (ICMP type:3, code:13, Communication
administratively prohibited)

PC1> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=37.358 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=21.574 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=31.881 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=28.448 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=37.021 ms

```

4.2 Extended (Named) ACL

What is an Extended ACL and why use it?

An Extended ACL is more powerful than a Standard ACL. It can check the source address, the destination address, and even the type of traffic (for example, only ping/ICMP, or only web traffic). This makes it possible to write much more precise rules. Giving the ACL a name instead of a number (a "Named" ACL) also makes it easier to read and manage, because the name describes what the rule is for.

Purpose in this assignment: the Finance department should be able to test connectivity to the IT department using ping (ICMP) only, for basic monitoring, but nothing else. The HR department should have no access to IT at all.

Configuration

```

ip access-list extended finance_to_it_icmp-only
permit icmp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
permit ip any any
interface FastEthernet1/0.30
ip access-group finance_to_it_icmp-only out

```

Note: the third line above (denying HR traffic to IT) was added after testing uncovered a gap in the original design. This is explained in full in section 6 (Troubleshooting Report).

Verification

Extended/Named ACL configuration on R1


```

R1#enable
R1#config te
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip access-list extended finance_to_it_icmp-only
R1(config-ext-nacl)#permit icmp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
R1(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
R1(config-ext-nacl)#permit ip any any
R1(config-ext-nacl)#exit

```

```

r1(config)#ip access-list extended finance_to_it_icmp-only
r1(config-ext-nacl)#25 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
r1(config-ext-nacl)#exit
r1(config)#exit

```

'show access-lists finance_to_it_icmp-only' showing match counters

```

r1#show access-lists finance_to_it_icmp-only
Extended IP access list finance_to_it_icmp-only
 10 permit icmp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 (5 matches)
 20 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 25 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255 (15 matches)
 30 permit ip any any (5 matches)
r1#

```

Insert Screenshot Here – PC2 (Finance) successfully pinging PC3 (IT)

```

PC2> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=45.679 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=39.370 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=34.472 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=32.884 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=22.459 ms

```

4.3 Named ACL for SSH Access

What is this ACL for?

This is another Named Extended ACL, but instead of controlling traffic between departments, it controls who is allowed to remotely log in to the router itself. It is applied to the router's VTY (virtual terminal) lines, which is where remote login sessions such as SSH come in.

Purpose in this assignment: only the IT department should be able to remotely manage the router using SSH. HR and Finance should not be able to reach the router's management session at all.

Configuration

```

ip access-list extended it-ssh-access
permit tcp 192.168.30.0 0.0.0.255 any eq 22
deny tcp any any eq 22
permit ip any any
line vty 0 4
access-class it-ssh-access in

```

Verification

Named ACL definition and 'show access-lists it-ssh-access' match counters

```
r1#show access-lists it-ssh-access
Extended IP access list it-ssh-access
 10 permit tcp 192.168.30.0 0.0.0.255 any eq 22 (6 matches)
 20 deny tcp any any eq 22
 30 permit ip any any
r1#show running-config | section it-ssh-access
ip access-list extended it-ssh-access
permit tcp 192.168.30.0 0.0.0.255 any eq 22
deny tcp any any eq 22
permit ip any any
access-class it-ssh-access in
```

ACL applied on VTY lines ('show running-config | section line vty')

```
r1#show running-config | section line vty
line vty 0 4
 access-class it-ssh-access in
 login local
 transport input ssh
r1#
```

4.4 SSH Secure Remote Management

What is SSH and why not Telnet?

Telnet is an older way to remotely log in to a network device, but it sends everything — including the password — as plain, readable text. Anyone who can see the traffic can read the password. SSH (Secure Shell) does the same job but encrypts the whole session, so passwords and commands cannot be read by anyone listening on the network. Because of this, SSH is the standard choice for managing real company devices.

Configuration

```
hostname r1
ip domain-name itsimplera.local
crypto key generate rsa modulus 2048
username admin privilege 15 secret Admin@123
line vty 0 4
 login local
 transport input ssh
ip ssh version 2
```

Verification

SSH access was tested from both an allowed source (IT) and a blocked source (HR), to confirm that both the SSH service and the Named ACL from section 4.3 are working correctly together.

SSH configuration on R1 ('show ip ssh' and related settings)

```

r1#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDQgxNSVJcBCdzVfzj3+yrsBhLsFZ8goP20xvz20izy
Wv6whkUnschHS9JlDR4caXJTGwa4gywPkKsA66PNX7jYuZReA0pVg1PI+9LCoq49aZZcM4MXtbH8Sk7j
f00/mmWGKRkT99N8HEdBKEAmNhZalqUurU53EolF2M0NqUcYJadgC8e9PJFia47m7PxXz5CD+P6JibH/
oUMfYfKx3lO6DIcIMqYkeUOJS6nTgT9I6YJz7FOTEzbp3nIA5/CnikxQblQSZx/fUp7FBHMC1YhawRU0
NAOkeP6Ey/RUn95fyLGSYKwce6Szu9u3k02XDEgEjKP6nqltRTb9HcuY4ulj
r1#show running-config | include hostname|domain|ssh|username
hostname r1
no ip domain lookup
ip domain name itsimplera.local
username admin privilege 15 secret 5 $1$0R55$XDGuDSYp8Se0jzLu.0e.x/
ip ssh version 2
ip access-list extended it-ssh-access
 access-class it-ssh-access in
 transport input ssh
r1#

```

Successful SSH login to R1 from the IT switch (SW2)

```

SW2>ssh -l admin 192.168.30.1
Password:

*Jul 25 06:01:06.004: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Gigabit
thernet0/0 (not half duplex), with r1.itsimplera.local FastEthernet1/0 (half duplex)
.r1#
r1#exit

[Connection to 192.168.30.1 closed by foreign host]
SW2>

```

SSH access denied when attempted from the HR switch (SW1)

```

SW1>ssh -l admin 192.168.30.1
% Destination unreachable; gateway or host down

SW1>

```

4.5 Port Security

What is Port Security and why use it?

Port Security is a switch feature that limits which device (by its MAC address) is allowed to use a specific switch port. Without it, if someone unplugs a PC and plugs in another device such as an unknown laptop or a small unmanaged switch — the network would simply let it connect. Port Security stops this by only allowing a fixed number of known MAC addresses on each port, and reacting if a different device tries to connect.

Configuration (applied to each PC-facing port on SW1 and SW2)

```

interface gigabitEthernet0/1
 switchport mode access
 switchport access vlan 10
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation restrict

```


The same pattern was repeated on every PC-facing port, using each port's correct VLAN (10 for HR, 20 for Finance, 30 for IT). "Sticky" MAC learning means the switch automatically learns and remembers the first MAC address it sees on the port, instead of it having to be typed in by hand. "Violation restrict" means that if a second, unknown device tries to use the same port, its traffic is silently dropped and a counter goes up, but the port itself is not shut down.

Verification

Port Security configuration on SW1 (Gi0/1, Gi0/2)

```
SW1#config ter
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#interface gigabitEthernet0/1
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security maximum 1
SW1(config-if)#switchport port-security mac-address sticky
SW1(config-if)#switchport port-security violation restrict
SW1(config-if)#exit
SW1(config)#interface gigabitEthernet0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 20
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security maximum 1
SW1(config-if)#switchport port-security mac-address sticky
SW1(config-if)#switchport port-security violation restrict
SW1(config-if)#exit
SW1(config)#exit
SW1#wr
*Jul 25 06:14:11.906: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
Compressed configuration from 3304 bytes to 1847 bytes[OK]
*Jul 25 06:14:17.789: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on
disk. Please wait...
*Jul 25 06:14:18.505: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk
successfully.
SW1#write memory
```

Port Security configuration on SW2 (Gi0/1, Gi0/2)

```
SW2#config ter
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#interface gigabitEthernet0/1
SW2(config-if)#switchport mode access
SW2(config-if)#
*Jul 25 06:13:51.448: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitE
thernet0/0 (not half duplex), with r1.itsimplera.local FastEthernet1/0 (half duplex)
.
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 30
SW2(config-if)#switchport port-security
SW2(config-if)#switchport port-security maximum 1
SW2(config-if)#switchport port-security mac-address sticky
SW2(config-if)#switchport port-security violation restrict
*Jul 25 06:14:44.598: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitE
thernet0/0 (not half duplex), with r1.itsimplera.local FastEthernet1/0 (half duplex)
.
SW2(config-if)#switchport port-security violation restrict
SW2(config-if)#exit
SW2(config)#interface gigabitEthernet0/2
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 30
SW2(config-if)#switchport port-security
SW2(config-if)#switchport port-security m
*Jul 25 06:15:30.248: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitE
thernet0/0 (not half duplex), with r1.itsimplera.local FastEthernet1/0 (half duplex)

SW2(config-if)#switchport port-security maximum 1
SW2(config-if)#switchport port-security mac-address sticky
SW2(config-if)#switchport port-security violation restrict
SW2(config-if)#exit
SW2(config)#exit
```

'show port-security interface' and 'show port-security address' on SW1

```
SW1#show port-security interface gigabitEthernet0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Restrict
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 0050.7966.6800:10
Security Violation Count : 0

SW1#show port-security address
Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
(mins)
-----
10      0050.7966.6800   SecureSticky        Gi0/1    -
20      0050.7966.6801   SecureSticky        Gi0/2    -
-----
Total Addresses in System (excluding one mac per port)  : 0
Max Addresses limit in System (excluding one mac per port) : 4096
SW1#
```

'show port-security interface' and 'show port-security address' on SW2

```
SW2#show port-security interface gigabitEthernet0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Restrict
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 0050.7966.6802:30
Security Violation Count : 0

SW2#show port-security address
Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
(mins)
-----
30      0050.7966.6802   SecureSticky        Gi0/1    -
30      0050.7966.6803   SecureSticky        Gi0/2    -
-----
Total Addresses in System (excluding one mac per port)  : 0
Max Addresses limit in System (excluding one mac per port) : 4096
SW2#
```

4.6 DHCP Snooping

What is DHCP Snooping and why use it?

Normally, computers get their IP address automatically from a DHCP server. DHCP Snooping is a switch feature that protects this process. It splits switch ports into two types: "trusted" ports, where real DHCP server replies are allowed to come from (normally the uplink towards the router or a real DHCP server), and "untrusted" ports, which are the normal PC-facing ports. If a DHCP server reply is ever seen coming from an untrusted port meaning someone connected a fake or rogue DHCP server the switch simply drops it, protecting the rest of the network from getting the wrong IP settings.

Configuration

```
ip dhcp snooping
ip dhcp snooping vlan 10,20
interface gigabitEthernet0/0
ip dhcp snooping trust
```

The same feature was enabled on SW2 for VLAN 30, trusting its uplink port towards R1. The PC-facing ports were left as untrusted, which is the correct and default behaviour.

Note: in this lab, the PCs were given fixed (static) IP addresses instead of using DHCP, so there was no live DHCP traffic during testing. Because of this, the DHCP Snooping binding table stays empty this is expected, and it does not mean the feature is not working. The trusted/untrusted boundary is still correctly configured and would apply automatically as soon as real DHCP traffic appeared on the network.

Verification

DHCP Snooping configuration on SW1 and SW2

```
SW1>enable
SW1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#ip dhcp snooping vlan 10,20
SW1(config)#interface gigabitEthernet0/0
SW1(config-if)#ip dhcp snooping trust
SW1(config-if)#exit
SW1(config)#interface gigabitEthernet0/1
SW1(config-if)#spanning-tree portfast
*Jul 25 12:28:41.735: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vios_12-adventerprisek9-m' passed code signing verification
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
SW1(config-if)#spanning-tree bpduguard enable
SW1(config-if)#exit
SW1(config)#interface gigabitEthernet0/2
SW1(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
SW1(config-if)#spanning-tree bpduguard enable
SW1(config-if)#exit
SW1(config)#exit

SW2>enable
SW2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#ip dhcp snooping
SW2(config)#ip dhcp snooping vlan 30
SW2(config)#interface gigabitEthernet0/0
SW2(config-if)#ip dhcp snooping trust
SW2(config-if)#exit
SW2(config)#interface gigabitEthernet0/1
SW2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
SW2(config-if)#spanning-tree bpduguard enable
SW2(config-if)#exit
SW2(config)#interface gigabitEthernet0/2
SW2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
SW2(config-if)#spanning-tree bpduguard enable
SW2(config-if)#exit
```

'show ip dhcp snooping' output on SW1 and SW2

```
SW1#show ip dhcp snooping
Switch DHCP snooping is enabled
Switch DHCP gleaning is disabled
DHCP snooping is configured on following VLANs:
10,20
DHCP snooping is operational on following VLANs:
10,20
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is enabled
  circuit-id default format: vlan-mod-port
  remote-id: 0cb6.61f0.0000 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:

Interface                Trusted    Allow option    Rate limit (pps)
-----
GigabitEthernet0/0       yes       yes             unlimited
  Custom circuit-ids:
SW1#show spanning-tree interface gigabitEthernet0/1 detail
Port 2 (GigabitEthernet0/1) of VLAN0010 is designated forwarding
  Port path cost 4, Port priority 128, Port Identifier 128.2.
  Designated root has priority 32778, address 0cb6.61f0.0000
  Designated bridge has priority 32778, address 0cb6.61f0.0000
  Designated port id is 128.2, designated path cost 0
  Timers: message age 0, forward delay 0, hold 0
  Number of transitions to forwarding state: 1
  The port is in the portfast edge mode
  Link type is point-to-point by default
  Bpdu guard is enabled
  BPDU: sent 287, received 0
```

```
SW2#show ip dhcp snooping
Switch DHCP snooping is enabled
Switch DHCP gleaning is disabled
DHCP snooping is configured on following VLANs:
30
DHCP snooping is operational on following VLANs:
30
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is enabled
  circuit-id default format: vlan-mod-port
  remote-id: 0c40.ec59.0000 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:

Interface                Trusted    Allow option    Rate limit (pps)
-----
GigabitEthernet0/0       yes       yes             unlimited
  Custom circuit-ids:
SW2#show spanning-tree interface gigabitEthernet0/1 detail
Port 2 (GigabitEthernet0/1) of VLAN0030 is designated forwarding
  Port path cost 4, Port priority 128, Port Identifier 128.2.
  Designated root has priority 32798, address 0c40.ec59.0000
  Designated bridge has priority 32798, address 0c40.ec59.0000
  Designated port id is 128.2, designated path cost 0
  Timers: message age 0, forward delay 0, hold 0
  Number of transitions to forwarding state: 1
  The port is in the portfast edge mode
  Link type is point-to-point by default
  Bpdu guard is enabled
  BPDU: sent 287, received 0
```

4.7 PortFast and BPDU Guard

What are PortFast and BPDU Guard, and why use them?

Normally, when a switch port comes up, Spanning Tree Protocol (STP) makes it wait through a few stages (listening, then learning) before it starts forwarding traffic. This delay, usually around 30 seconds, exists to prevent loops, but it is unnecessary and annoying on a port that is simply connected to a PC, since a PC cannot cause a switching loop. PortFast tells the switch to skip straight to forwarding on these ports.

The trade-off is that PortFast assumes nothing but a PC will ever be connected to that port. BPDU Guard is the safety net for that assumption: if a PortFast port ever receives a BPDU (a special Spanning Tree control

message, which only switches send), it means someone has plugged in another switch by mistake or on purpose. BPDU Guard immediately disables that port to protect the network from a possible loop.

Configuration (applied to each PC-facing port on SW1 and SW2)

```
interface gigabitEthernet0/1
spanning-tree portfast
spanning-tree bpduguard enable
```

The same two commands were applied to every PC-facing port on both switches (Gi0/1 and Gi0/2 on SW1 and SW2).

Verification

'show spanning-tree interface ... detail' confirming PortFast and BPDU Guard

```
SW1#show spanning-tree interface gigabitEthernet0/1 detail
Port 2 (GigabitEthernet0/1) of VLAN0010 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.2.
Designated root has priority 32778, address 0cb6.61f0.0000
Designated bridge has priority 32778, address 0cb6.61f0.0000
Designated port id is 128.2, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast edge mode
Link type is point-to-point by default
Bpdu guard is enabled
BPDU: sent 287, received 0
```

```
SW2#show spanning-tree interface gigabitEthernet0/1 detail
Port 2 (GigabitEthernet0/1) of VLAN0030 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.2.
Designated root has priority 32798, address 0c40.ec59.0000
Designated bridge has priority 32798, address 0c40.ec59.0000
Designated port id is 128.2, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast edge mode
Link type is point-to-point by default
Bpdu guard is enabled
BPDU: sent 287, received 0
```

4.8 Root Guard

What is Root Guard and why use it?

Every switched network running Spanning Tree Protocol elects one switch as the "root bridge", which becomes the reference point the whole tree is built around. Normally this election happens automatically, and network administrators design the network so that a specific, reliable core switch is meant to win that election. Root Guard is placed on a port to guarantee that no switch connected through that port is ever allowed to become the root bridge — even if it advertises information that would normally win the election. This protects the network from a wrongly configured switch, or a malicious one, taking over as the root and changing how traffic flows.

In this lab's original design, SW1 and SW2 were only connected through router R1, with no direct switch-to-switch link, so there was no natural place to demonstrate Root Guard. To properly test this feature, an extra link was added directly between SW1 (Gi0/3) and SW2 (Gi0/3), configured as a trunk carrying only VLAN 1, and Root Guard was enabled on both ends.

Configuration

```
interface gigabitEthernet0/3
switchport trunk encapsulation dot1q
switchport mode trunk
```



```
switchport trunk allowed vlan 1
spanning-tree guard root
```

The same configuration was applied on the matching Gi0/3 interface on the other switch.

Testing performed

- SW2's Spanning Tree priority was temporarily set to 0 (the lowest possible value) to simulate a switch trying to force itself to become the root bridge.
- SW1's Root-Guard-protected port (Gi0/3) immediately detected this and placed itself into a blocking "Root Inconsistent" state, refusing to let SW2 become the root through that link.
- SW2's priority was then set back to its default value to remove the simulated attack.
- The port on SW1 remained blocked even after the priority was reverted. This was investigated and explained in section 6 (Troubleshooting Report) — SW1 was then given a permanently lower priority so that it is always the intended root bridge, after which the port recovered normally.

Verification

Root Guard configuration on SW1 and SW2 (Gi0/3)

```
SW1>enable
SW1#config ter
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#interface gigabitEthernet0/3
SW1(config-if)#switchport trunk encapsulation dot1q
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk allowed vlan 1
SW1(config-if)#spanning-tree guard root
SW1(config-if)#
*Jul 25 07:11:35.568: %SPANTREE-2-ROOTGUARD_CONFIG_CHANGE: Root guard enabled on port
GigabitEthernet0/3.
*Jul 25 07:11:35.981: %SPANTREE-2-ROOTGUARD_BLOCK: Root guard blocking port GigabitE
thernet0/3 on VLAN0001.
SW1(config-if)#exit
SW1(config)#exit

SW2>enable
SW2#config ter
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#interface gigabitEthernet0/3
SW2(config-if)#switchport trunk encapsulation dot1q
SW2(config-if)#switchport mode trunk
SW2(config-if)#switchport trunk allowed vlan 1
SW2(config-if)#spanning-tree guard root
SW2(config-if)#exit
*Jul 25 07:09:41.611: %SPANTREE-2-ROOTGUARD_CONFIG_CHANGE: Root guard enabled on port
GigabitEthernet0/3.
SW2(config)#exit
```

show spanning-tree inconsistentports' — port blocked during the simulated attack

```
SW2>enable
SW2#config ter
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#spanning-tree vlan 1 priority 0
SW2(config)#exit
SW2#
```

SW1 given permanent lower priority to become the intended root bridge

```
SW1#config ter
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#spanning-tree vlan 1 priority 4096
SW1(config)#exit
SW1#
```

'show spanning-tree inconsistentports' — port recovered

```
SW1>enable
SW1#show spanning-tree inconsistentports

Name                Interface          Inconsistency
-----
Number of inconsistent ports (segments) in the system : 0
```

5. Connectivity Test Results

After all the security features above were configured, the following end-to-end tests were carried out to confirm that the network behaves exactly as designed allowing the traffic that should be allowed, and blocking the traffic that should be blocked.

5.1 PC1 (HR) Inter-VLAN and ACL Enforcement Test

- PC1 → Gateway (192.168.10.1): success confirms basic connectivity within HR.
- PC1 → PC2 (Finance, 192.168.20.10): blocked confirms the Standard ACL (section 4.1) is working.
- PC1 → PC3 (IT, 192.168.30.10): blocked confirms the Extended ACL fix (section 4.2) is working.

PC1 ping results to gateway, Finance, and IT

```
PC1> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=25.140 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=12.071 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=8.077 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=19.665 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=13.063 ms

PC1> ping 192.168.20.10

192.168.20.10 icmp_seq=1 timeout
*192.168.10.1 icmp_seq=2 ttl=255 time=13.426 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.10.1 icmp_seq=3 ttl=255 time=8.629 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.10.1 icmp_seq=4 ttl=255 time=6.053 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.10.1 icmp_seq=5 ttl=255 time=9.696 ms (ICMP type:3, code:13, Communication administratively prohibited)

PC1> ping 192.168.30.10

192.168.30.10 icmp_seq=1 timeout
*192.168.10.1 icmp_seq=2 ttl=255 time=17.511 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.10.1 icmp_seq=3 ttl=255 time=5.462 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.10.1 icmp_seq=4 ttl=255 time=7.436 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.10.1 icmp_seq=5 ttl=255 time=15.821 ms (ICMP type:3, code:13, Communication administratively prohibited)
```

5.2 PC2 (Finance) ICMP-only Access to IT

- PC2 → PC3 (IT, 192.168.30.10): success — Finance is allowed ping-only access to IT, as designed.

PC2 successfully pinging PC3

```
PC2> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=45.679 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=39.370 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=34.472 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=32.884 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=22.459 ms
```

5.3 PC4 and PC3 (IT) Intra-VLAN Switching Test

PC3 and PC4 both belong to VLAN 30 and are connected to the same switch (SW2). Traffic between them should be switched locally and should never need to pass through router R1.

- PC4 → PC3 (192.168.30.10): success, with a low, consistent round-trip time confirming local (Layer 2) switching rather than routing.

PC4 pinging PC3, confirming intra-VLAN switching

```
PC4> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=64 time=12.758 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=64 time=3.292 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=64 time=5.809 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=64 time=3.767 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=64 time=5.682 ms
```

5.4 SSH Access Test

This test is already covered in detail in section 4.4 SSH succeeded from the IT department and was correctly denied from HR.

6. Troubleshooting Report

This section documents the real problems that came up while building and testing this network, and the steps taken to find and fix each one. These are included because troubleshooting skills are just as important as configuration skills in real network administration work.

6.1 GNS3 Virtual Link Desynchronization

Scenario

After the GNS3 lab had been running for a while, a simple ping test between SW1 and R1 suddenly returned 0% success, even though every relevant configuration VLANs, trunk settings, IP addressing, and ACLs was checked and found to be correct.

Observed Behaviour

A closer investigation using CDP (Cisco Discovery Protocol) showed zero neighbours on either device, even though CDP was enabled on both. A Wireshark packet capture placed directly on the R1–SW1 link showed traffic leaving R1, but nothing at all arriving from SW1's side even while a ping was actively being sent.

Resolution

This pointed to the virtual link inside GNS3 itself being out of sync with the running devices, rather than a configuration problem. Reloading the GNS3 project resynchronized the virtual link, and traffic began flowing normally again.

6.2 Duplex Mismatch Between R1 and Both Switches

Scenario

Even after the link above came back up, ping tests were not reaching 100% success, and later SSH sessions were seen to disconnect unexpectedly.

Observed Behaviour

CDP raised duplex mismatch warnings on both router-to-switch links: R1's FastEthernet interfaces were fixed at half duplex, while both switches were set to auto duplex, which had negotiated to full duplex on their side.

Resolution

Duplex was explicitly set to full on both of R1's interfaces (f0/0 and f1/0) to match the switches. Ping success immediately improved to 100%, and SSH sessions no longer disconnected mid-session.

6.3 SSH Cipher Mismatch

Scenario

SSH connections from the IT switch (SW2) to R1 were reaching the password prompt but then aborting immediately afterwards, before a session could be established.

Observed Behaviour

Turning on SSH debugging (debug ip ssh) on R1 showed a clear error: "No matching cipher found." The switch's SSH client was only offering modern AES-CTR ciphers, while R1's older IOS image only supported the older AES-CBC and 3DES-CBC ciphers, so the two sides could not agree on how to encrypt the session.

Resolution

The switch's SSH client was configured to also offer the CBC cipher suite that R1 supports. Once both sides had at least one cipher in common, the SSH handshake completed successfully and login worked as expected.

6.4 ACL Logic Gap HR Traffic Unexpectedly Reaching IT

Scenario

While testing intra-department connectivity, PC4 (IT) was able to successfully ping PC1 (HR), even though HR was never meant to have any access to the IT department at all.

Observed Behaviour

Checking the Extended ACL (finance_to_it_icmp-only) showed the reason: its final line was a catch-all "permit ip any any", added so that normal, unrelated traffic would still be allowed through. This line was written broadly enough that it also allowed HR traffic to IT, since only Finance traffic had an explicit rule restricting it. HR traffic was never mentioned in the ACL at all, so it fell through to the catch-all permit.

Resolution

An explicit rule was added to the ACL, denying all IP traffic from the HR subnet to the IT subnet. This closed the gap completely while leaving Finance's ICMP-only access, and all other unrelated traffic, working exactly as before.

6.5 Root Guard Port Stayed Blocked After Removing the Simulated Attack

Scenario

After testing Root Guard by temporarily lowering SW2's Spanning Tree priority to simulate an attack, the priority was set back to its normal, default value. However, SW1's protected port stayed in the blocked "Root Inconsistent" state instead of recovering.

Observed Behaviour

Checking Spanning Tree details showed that, at equal (default) priority, SW2's built-in MAC address happened to be numerically lower than SW1's. Because Spanning Tree uses the lowest MAC address as a tie-breaker when priorities are equal, SW2 was still technically the "better" candidate to be root so Root Guard kept blocking the port, exactly as it is supposed to.

Resolution

SW1 was given a permanently lower Spanning Tree priority than the default, making it the clearly intended root bridge regardless of MAC address. Once SW2 could no longer look better than SW1 by any measure, the blocked port on SW1 automatically recovered to a normal forwarding state.

7. Challenges Faced

- The GNS3 virtual link between devices went out of sync after the lab had been running for a while, causing a complete loss of connectivity that was not visible from the configuration alone and needed a project reload to fix.

- Cisco IOS interface naming is different between device types FastEthernet on the router versus GigabitEthernet on the switches which had to be checked carefully with 'show ip interface brief' before applying any interface-level commands.
- An IOU-format switch image was briefly considered as an extra device, but it turned out to be incompatible with the QEMU-based nodes already used in this lab, so the original topology and device types were kept instead.
- Finding the exact place where an ACL's catch-all rule could accidentally allow unwanted traffic required careful, step-by-step testing between every pair of departments, rather than assuming the configuration was correct just because it matched the written design.
- Demonstrating Root Guard meaningfully required adding an extra direct link between the two switches, since the original topology only connected them through the router.

8. Conclusion

This assignment provided hands-on experience in securing a small enterprise network. Access Control Lists Standard, Extended, and Named were used together to enforce a clear department-by-department security policy: HR fully isolated from Finance and IT, Finance limited to ping-only monitoring access to IT, and only the IT department allowed to remotely manage the router over SSH.

Layer 2 security features (Port Security, DHCP Snooping, BPDU Guard, Root Guard, and PortFast) were then added on top of this to protect the switches themselves from common mistakes and simple attacks, such as an unknown device being plugged in or a switch wrongly trying to become the network's root bridge.

Several real problems came up during this assignment a desynchronized virtual link, a duplex mismatch, an SSH cipher mismatch, a gap in ACL logic, and a Spanning Tree tie-breaker behaviour and each one was diagnosed step by step and resolved rather than worked around. This troubleshooting process reinforced practical, real-world network administration skills that go beyond simply typing in configuration commands.